

Exercise set 4

Regression Trees

Task 1

- a) When we fit a regression tree, we sequentially divide the predictor space by finding the variable X_j and the cutpoint s that reduces the RSS as much as possible. Predict the natural logarithm of **Salary** using **Hits** and **Years** in the **Hitters** data. The code below is doing the first split in the regression tree. Explain what the code does. Use the help functions in R if necessary.

```
library(ISLR)
RSS=function(s,x1,y)
{
  yR1=mean(y[x1<s])
  yR2=mean(y[x1>=s])
  rss=sum((y[x1<s]-yR1)^2)+sum((y[x1>=s]-yR2)^2)
  return(rss)
}
Hitters=Hitters[!is.na(Hitters$Salary),]
Hitters$logSalary=log(Hitters$Salary)
optimise(RSS,lower=0,upper=250,x1=Hitters$Hits,y=Hitters$logSalary)

## $minimum
## [1] 117.5044
##
## $objective
## [1] 160.9715

optimise(RSS,lower=0,upper=50,x1=Hitters$Years,y=Hitters$logSalary)

## $minimum
## [1] 4.508575
##
## $objective
## [1] 115.0585
```

- b) Use the **tree**-function in the **tree**-package to fit and plot a tree for **logSalary** and the variables **Hits** and **Years** only.

```
library(tree)
tree1=tree(logSalary~Hits+Years,data=Hitters)
plot(tree1)
text(tree1,pretty=0)
```

Can you see where your computations in a) occur in the tree?

- c) Predict the salary of a new observation with **Hits**=125 and **Years**=2.5.
- d) Fit a tree for **logSalary** with all predictors; call this **tree2**. Compute the RSS for **tree1** and **tree2**. Also, use the summary function to see if you can find RSS there.

Bagged trees and random forest

Task 2

Bagging is based on the idea of reducing prediction variance by averaging over predictions for many bootstrapped samples.

- Use a cross-validation approach to compute the `testMSE` for the `tree2` in Task 1. We are going to use this as a benchmark to compare with bagging, random forest and boosting.
- Use bagging of `tree2` in Task 1 to see if you can improve the predictions. Use the `randomForest` package. Use the variable importance plot, `varImpPlot` and interpret the results. What is the unit of the x-axis of the plot. Use the help-function to the `importance`-function to find this out.
- Can you improve the predictions with random forest? You have to use the `randomForest`-function in a modified way compared to in b). Random forest is an extension of bagging. In what sense? Why are random forests potentially better predictors?

Boosting

In boosting, predictions are also averaged, just like in bagging and random forest. However, the dataset is updated sequentially so that residuals from the previous iteration is used as data in the next.

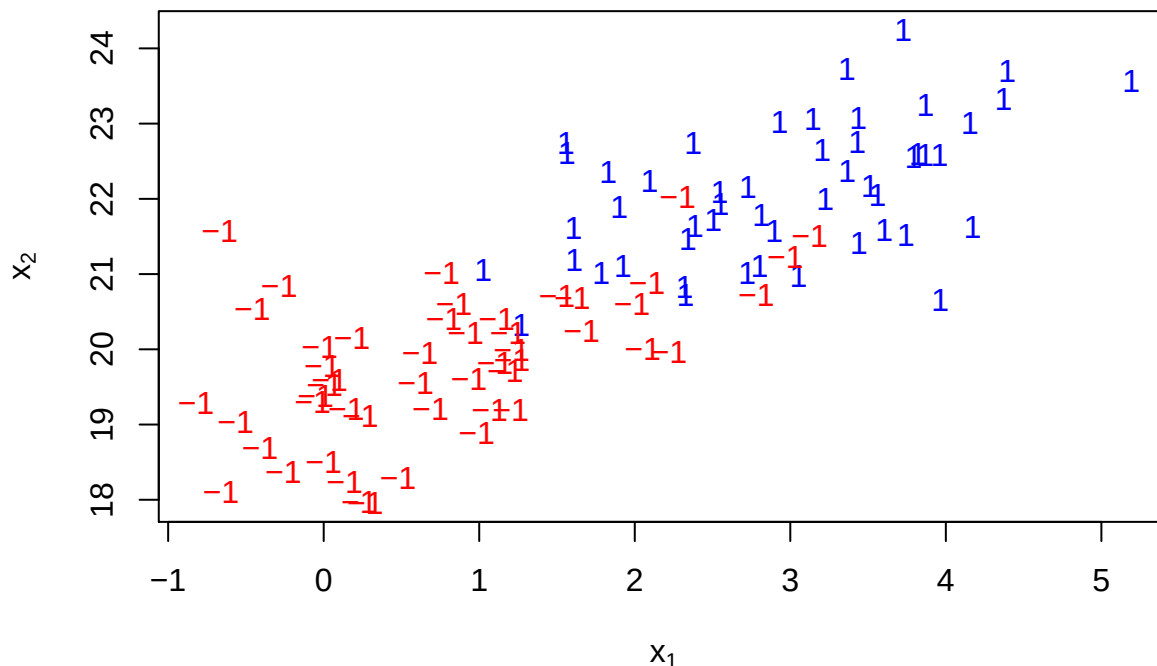
Task 3

Again, you shall predict `logSalary` in the `Hitters` data. Use the `gbm`-function in the `gbm`-package to predict with a boosted tree. Compute `testMSE`.

Support vector machines

Task 4

- In the plot below where the classes “1” and “-1” are denoted, is it possible to find a maximal marginal classifier?



- b) The dataset in a) can be downloaded as the file “xy.csv” in Canvas. Download it, read it into R and recreate the plot in a).
- c) Use the support vector classifier to classify the dataset. See lecture13.pdf for how to do this.
- d) What does it mean to set the tuning parameter C equal to 10?
- e) Try a support vector machine with a polynomial kernel and check if it is able to separate the training data better?

Principal component analysis (PCA)

Task 5

- a) Use principal component analysis to analyze the **advertising** data in “advertising.csv”, which you will find in Canvas. Find the principal first two principal components for the variables **TV**, **radio** and **newspaper**. Interpret them by looking at the loadings.

Cluster analysis

Task 6

- a) Use K-means clustering to cluster the states in the USArrest data in three different groups based on the variables **Murder** and **Assault**.
- b) Use hierarchical clustering.